# Research Proposal: Topics in Econometrics

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## 1 Introduction

In many education systems, students must make binding track choices at a very young age. These decisions occur under severe uncertainty about ability, preferences, and future success (Betts, 2011; Dustmann et al., 2012; Giustinelli and Pavoni, 2017), and they are largely irreversible: early tracking placements strongly shape higher education access, labor market trajectories, and the dynamics of skill accumulation (Hanushek and Woessmann, 2006; Dustmann et al., 2012; De Groote, 2025; Humphries et al., 2025). The information students have when making these choices is therefore consequential — and where that information comes from matters.

A large literature documents the rigidities and inequalities generated by tracking, but the "soft" institutional mechanisms governing these choices remain largely unexplored. In many OECD countries — including Germany, Belgium, the Netherlands, Switzerland, and Austria — schools issue formal track recommendations that act as powerful signals at a stage when students are actively forming beliefs about their own comparative advantage (Piopiunik, 2014; Papageorge et al., 2020; Giustinelli and Pavoni, 2017). This advice embeds soft information — effort, relative class rank, classroom behavior — that goes beyond objective test scores, and it is susceptible to institutional bias (Carlana, 2019; Brunello et al., 2025). Whether this advice helps students or distorts their choices depends on what schools know, what schools want, and how students update beliefs in response.

A natural framework for thinking about this is the growing structural literature on educational choice under information frictions (Stinebrickner and Stinebrickner, 2012, 2014; Larroucau and Rios, 2024; Arcidiacono et al., 2025). In this literature, students observe noisy signals — grades, wages, output — and update Bayesianly about their latent ability. But signals are treated as features of the environment: they arrive *exogenously*. Many real institutional signals are not like this. They are issued by strategic actors with their own objectives: teachers, supervisors, lenders, doctors. Modeling these signals as exogenous misses what may be the central feature of how information flows through educational institutions — that the signal a student receives reflects both what the institution knows and what the institution wants.

This project models educational choice under uncertainty when one of the signals is endogenously produced by an institution with conflicting interests. Schools in our setting play a dual role. They observe more than students do — relative class rank, behavioral cues, accumulated experience mapping past cohorts to outcomes — which makes their recommendations potentially informative. But they also face capacity constraints and outcome-reporting incentives that may not align with student welfare, which means their recommendations are also shaped by institutional pressures unrelated to what is best for the student. A school may, in equilibrium, push high-ability students into lower tracks to manage cohort sizes. Whether this happens, how it varies across school types, and what it costs students are the empirical questions this paper addresses. Students, on the other side, do not receive this advice passively: they are Bayesians who combine the school's signal with public information (grades) and their own private knowledge, with response heterogeneity moderated by family background.

Belgium provides a clean setting to study this. At age 14, students commit to academic, technical, or vocational tracks; schools issue formal recommendations annually; and the Belgian LOSO dataset (De Groote, 2025) observes the full information set schools use to recommend (relative grade rank, relative effort rank), the standardized exams that form a balanced grade panel, and a dimension of plausibly different school behavior: cycle architecture. Some schools offer only the first cycle of secondary education, while others offer all three. Schools facing different cohort flows have plausibly different incentives in how they recommend — a source of variation we exploit for the counterfactual.

The project aims to make two contributions. First, it extends the structural learning literature (Stinebrickner and Stinebrickner, 2012; Arcidiacono et al., 2025; Larroucau and Rios, 2024; Bunting et al., 2024) to settings where one of the signals is generated by a strategic institutional actor rather than arriving exogenously, providing the first such model applied to secondary-school tracking. Second, it provides empirical evidence on how heterogeneity in school recommendation rules — driven by institutional architecture — affects student welfare and the SES gradient in track outcomes, contributing to literatures on teacher expectations and the role of soft signals in tracking systems (Piopiunik, 2014; Carlana, 2019; Brunello et al., 2025).

## 1.1 Institutional Setting

Belgium has a secondary education system that implements tracking at a very early age. At 12, students enter high-school and must decide between two general tracks — denominated *A-stroom* and *B-stroom*. These two tracks, although general, already generate some classification: *A-stroom* prepares for academic and technical tracks, while *B-stroom* leads to vocational tracks.[1] The decision to enter A-stroom or B-stroom at grade 1 of high-school is generally unconstrained, although it is informed by previous school grades and advice from school teachers.

At age 14, students make a binding track choice. The options are: the academic track (**ASO**), the technical track (**TSO**), and the vocational track (**BSO**). The first is the most common option and leads mostly to university and professional colleges; the second is the least common and leads mostly to professional college; the third is the second most common and leads to professional training. Up to high-school graduation there is little upward mobility (BSO $\rightarrow$ TSO $\rightarrow$ ASO) but substantial downward mobility. Students may either learn that their skill does not match their track or be forced to downgrade if they fail. At the end of each year, students are issued a general certificate of performance: the A certificate allows students to continue and pick whatever option they want (upgrade, downgrade, or continue), the C certificate forces students to repeat track, and the B certificate forces them to either downgrade or repeat the course in the same track.

Within ASO and TSO, students at age 14 also choose a subtrack tailored to their interests — e.g., "Latin and math" or "social sciences." The subtrack is highly predictive of the field they will enroll in college. Although entry to college is not restricted by entry exams and students are free to choose any field,[2] students risk skill mismatch and dropout. Accumulated skill in a subtrack during high-school is thus important for a successful higher-education path.

A key institutional feature of the Flemish system is the role schools play in these decisions. At the end of each high-school year, the school issues an official recommendation about which track and subtrack a student should be studying next year. These recommendations are part of the academic evaluation — which also includes the official certificate — and are formally communicated to families in June.[3]

[1] The dataset I was given by Olivier De Groote, used in De Groote (2025), classified students from *A-stroom* and *B-stroom* into the tracks that later will be binding. This was done based on the track — A leads to TSO and ASO and B to BSO — and on the electives picked, which were used to distinguish ASO from TSO. For more information see Appendix B of De Groote (2025).

[2] University is mostly public and has very low fees.

[3] The recommendation is formally issued by the class teacher (*Klasstitularis*) after consultation with colleagues. We refer throughout to "schools" as the issuer for generality, but the institutional reality is teacher-based. We treat the

The decision at age 14 is consequential because it can lock students in certain paths. Downward mobility is possible based on certificates and performance, but upward mobility is rare due to skill accumulation; horizontal mobility is possible but movement from non-STEM to STEM is rarely observed. In the following years, schools again re-recommend and issue certificates, but the lock-in effects of the age-14 choice make subsequent recommendations less consequential.

A second key feature is institutional variation across schools. Schools differ in two dimensions: which tracks they offer (ASO, TSO-BSO, or all together — roughly 33% each) and which cycles they offer (some only cycle 1, grades 1-2; others cycles 1, 2, and 3). The latter — cycle architecture — generates plausibly different incentives for school recommendation behavior, and forms one of the central sources of identifying variation for the counterfactuals I have in mind.

## 1.2 Data

The LOSO dataset follows a cohort of ≈6,400 students from 89 high-schools in the Flemish region of Belgium, collected during the 1990s. The dataset tracks students from age 12 through higher-education entry — approximately 8.5 years.[4] Here I provide a summary of the available information, at both the student and the school level:

**Student-side variables.**

- **Track choices.** For each year, we observe the track-subtrack chosen, the certificate received, and dropout. Attrition is minimal. The first two years of higher education are also observed, including type of college, degree etc.
- **Grades.** Observed for grades 1, 2, 4, and 6 in both Dutch and Math. Crucially, these grades are administered in a standardized format across the entire sample, independently of the track or subtrack the student is in. This balanced panel of comparable performance measures is what enables the AAMR-style identification of learning parameters without selection correction (§2).
- **Baseline measures.** Cognitive skills, non-cognitive skills, and parental involvement, observed at $t = 0$. Used for identification of unobserved types via Hu–Schennach measurement system (§1).
- **Subjective beliefs.** For each year corresponding to grades observation, we observe the track-subtrack the student expects to graduate in (i.e "ASO-STEM"). From grade 4 onwards, we also observe the job or higher-education path the student expects. These reported expectations provide direct moments on student beliefs over future track outcomes and can be used in identification.
- **Location.** Student's (home) location, allowing computation of distance to school as exogenous variation.

**School-side variables.**

- **Type of school.** Tracks offered and cycles offered (cycle architecture). Roughly 33% of schools fall into each track-offering category. Cycle architecture (C1 vs C123) is the central source of variation for the counterfactual.

recommendation as the equilibrium outcome of within-school deliberation and informal communication with parents during the year; identifying parental influence as a separate structural object is beyond the scope of this paper. CF, however, need to avoid changing information structure to survive Lucas Critique.

[4] Schools were selected from two regions: Dendermonde-Zele-Hamme and Diest-Hasselt-Genk-Houthalen-Leopoldsburg-Beringen, based on criteria such as school size, administrative structure, and representation of different school types (public and private education).

- **Capacity constraints.** Number of seats per grade and track. Reported at the beginning of the panel based on the population, not the sampled students.
- **General school statistics:** Also given at the beginning of the panel and for other students, schools also provide the percentage of dropout, the percentage of students going to H.E etc (success measures).
- **Recommendations.** The track-subtrack recommended to each student for different periods in time. We collapse the 20 official ASO subtracks (same for TSO) into STEM/non-STEM categories for tractability; alternative taxonomies are possible, but the literature agrees on checking this dimension (Arcidiacono et al., 2025; Humphries et al., 2025).
- **School information set.** For each class, we observe how the school rates the student in the within-class performance distribution in Math, Dutch, both for exams and effort given. These rankings form the basis of the school's information set $Z$ in the model (§2).
- **Advice process variation.** We additionally observe survey responses on how each school structures its advice process — whether it holds parent meetings during the year, whether it consults students before issuing recommendations, and what information about future tracks it provides. This information is informative about residual school heterogeneity beyond cycle architecture and can be used in robustness analyses.

## 2 Model sketch

I present the sketch of the model. First, I define the timing and the players. Second, I describe the sources of unobserved heterogeneity, the key primitives, and the information structure. Then the production functions and the school side. Estimation details go to the appendix and are mostly sketched for brevity.

### 2.1 Environment and timing

**Time.** Discrete annual periods $t \in \{0, 1, 2, 3, \dots, T\}$:

- $t = 0$: start of cycle 1. Baseline measurements $M_i$ taken; school enrollment $s_i$ realized.
- $t \in \{1, 2\}$: cycle 1. End-of-year standardized exams $G_{it}$ administered; school-issued recommendation $R_{it}$ delivered after grades.
- $t = 3$: binding initial track choice $d_{i3}$ at start of cycle 2.
- $t = 4, 5$: cycle 2 and 3. Annual track-refinement decisions $d_{it}$ allowed within a downgrade trajectory. Again $R_{it}$, $G_{it}$ are realized.
- $t = T$: terminal outcome $Y_i^{\text{post}}$ (HE entry / labour-market signal).

Track movements are restricted: BSO is absorbing[5], and within-cycle moves follow the downgrade order ASO $\to$ TSO $\to$ BSO. Upgrades are not feasible, but movements on the STEM/non-STEM margin are allowed.

**Players.** Students $i = 1, \dots, N$; schools $s \in \mathcal{S}$, indexed by cycle-architecture type $\tau(s) \in \{\text{C1}, \text{C123}\}$ (cycle-1-only vs. full cycle). Track set is defined as:

$$\mathcal{J} = \{\text{ASO-S},\ \text{ASO-N},\ \text{TSO-S},\ \text{TSO-N},\ \text{BSO}\}, \qquad |\mathcal{J}| = 5. \tag{1}$$

[5]Justified by the low rate of upgrades from BSO in the data.

## 2.2 Sources of heterogeneity

The model has three sources of unobserved heterogeneity: $\theta_i^k$, a discrete type known to student and school but not to the econometrician; $\theta_i^u$, a track-specific match quality unknown to all agents at $t = 0$; and the different idiosyncratic shocks $\varepsilon_{it}$. This setting takes mostly as reference the notation and setting developed in Arcidiacono et al. (2025); Bunting et al. (2024).

**Known type $\theta_i^k$.** Discrete:

$$\theta_i^k \in \{1, \ldots, K\}, \qquad \Pr(\theta_i^k = k) = \pi_k, \quad \sum_k \pi_k = 1. \tag{2}$$

Each type $k$ encodes a profile over four latent dimensions: cognitive skills, a non-cognitive component, and a family-pressure component, $(\theta^k_{\text{num},k}, \theta^k_{\text{verb},k}, \theta^k_{\text{noncog},k}, \theta^k_{\text{fam},k})$. Identification follows Humphries et al. (2025) and Arcidiacono et al. (2025): I use selection-free baseline measurements with at least three repeated measures per dimension[6]. Table 1 in the appendix maps observed measures to latent factors.

**Unknown type $\theta_i^u$.** Track-specific match quality, unknown to all agents at $t = 0$:

$$\theta_i^u = (\theta_{i,j}^u)_{j \in \mathcal{J}} \in \mathbb{R}^5. \tag{3}$$

Following Arcidiacono et al. (2025), this is the latent component of the outcome the student would experience in track $j$, net of $\theta^k$ and $X$. It is the object students learn about over the panel; it absorbs effort and own realizations of skills within each of the tracks.

Conditional on $\theta^k$, the prior on $\theta^u$ is Gaussian:

$$\theta_i^u \mid \theta_i^k = k \;\sim\; \mathcal{N}\big(m_0(k),\ \Sigma_0\big), \tag{4}$$

with $\Sigma_0$ positive definite and *not* restricted to be diagonal (correlated learning across tracks, as in AAMR). The mean $m_0(k)$ is type-specific[7].

**Observed covariates.** $X_{it}$ are exogenous observables (gender, age, grade level).

Here I outline the two main assumptions on the initial distribution of skills:

**Assumption A1 (sorting on observable types).** Initial school assignment $s_i$ may depend on $\theta_i^k$ but $\theta_i^u \perp s_i \mid \theta_i^k$. This is justified because at $t = 0$ uncertainty over $\theta^u$ has not been realized.[8]

**Assumption A2 (Gaussian conjugacy on grades).** Grade noise is Gaussian and i.i.d. across $t$, conditional on $(\theta^k, \theta^u, X)$. This delivers closed-form Gaussian posteriors after grade signals; recommendation updating requires an additional approximation (A8 below). This is a standard assumption in the learning literature (Bunting et al., 2024).

[6] Identification additionally requires that the measurement loadings on latent factors are linearly independent — i.e., that no measure is a perfect linear combination of others within a factor.

[7] Potentially, it can also be $s$ specific, as students may enter with different confidence in each school.

[8] Priors over $\theta^u$ may differ across schools (e.g. if students hold more positive expectations in some schools). Testable by regressing $\hat{m}^G_{i,1}$ on school dummies controlling for $\theta^k$. If significant, $m_0$ is made school-specific.

## 2.3 Grade equation

The production function is described next.

For $t \in \{1,2,3,4\}$ and subject $b \in \{\text{Math}, \text{Dutch}\}$ we define:

$$G_{ibt} \;=\; \mu_b(\theta_i^k, X_{it}) \;+\; \Lambda_b^\top \theta_i^u \;+\; \varepsilon_{ibt}, \qquad \varepsilon_{ibt} \sim \mathcal{N}(0, \sigma_b^2),\ \text{i.i.d.} \tag{5}$$

- $\mu_b(\theta_i^k, X_{it})$: type-specific intercept plus linear in $X_{it}$.
- $\Lambda_b \in \mathbb{R}^5$: loading vector. Math grades load on STEM components, Dutch on non-STEM. Normalize one entry of $\Lambda_b$ per subject for scale.

A useful difference from Arcidiacono et al. (2025) is that $G_{ibt}$ are standardized exams observed for every student at every $t$, regardless of school or future track. Their grade likelihood requires participation indicators because they only observe grades for the major chosen. Our case does not:

$$L_i^G \;=\; \prod_{t=1}^{4} \prod_{b \in \{\text{Math},\text{Dutch}\}} \phi\big(G_{ibt};\, \mu_b(\theta_i^k, X_{it}) + \Lambda_b^\top \theta_i^u,\, \sigma_b^2\big). \tag{6}$$

The balanced grade panel sharply identifies $(\Lambda_b, \sigma_b^2, \Sigma_0)$ via cross-individual covariances, without the joint outcome-and-choice likelihood Arcidiacono et al. (2025) require.

## 2.4 School's information set and the DGP for $Z$

One of the key assumptions of the paper is that the school observes more than the student In addition to $(\theta_i^k, G_{it}, X_{it})$, it observes

$$Z_{it} \;=\; (\text{rank}_{it}^{Gb},\ \text{rank}_{it}^{Eb})_{b \in \{\text{Math},\text{Dutch}\}}, \tag{7}$$

where $\text{rank}_{it}^{Gb}$ is class-relative rank in grades for subject $b$ and $\text{rank}_{it}^{Eb}$ is class-relative rank in effort for subject $b$. Importantly, we also observe these variables that schools use to make decisions.

I model these rankings as discretized realizations of a latent school-side signal:

$$Z_{ibt}^* \;=\; \mu_b(\theta_i^k, X_{it}) + \lambda_b^\top \theta_i^u + \xi_{ibt}, \qquad \xi_{ibt} \sim \mathcal{N}(0,1), \tag{8}$$

with the observed category generated by thresholds:

$$Z_{ibt} = m \quad \Longleftrightarrow \quad \kappa_{m-1,b} < Z_{ibt}^* \le \kappa_{m,b}, \tag{9}$$

for $-\infty = \kappa_{0,b} < \kappa_{1,b} < \cdots < \kappa_{M_b,b} = +\infty$. This is standard ordered probit and is useful because it simplifies a bit the procedure we will use to turn the discrete signals into normality.

Importantly, $Z_{ibt}^*$ has the same structural form as grades (5) but with independent noise $\xi_{ibt}$, providing an additional draw on the same underlying $\theta^u$. The recommendation is informative beyond grades because $\xi$ is independent of $\varepsilon$. We can be more precise with the following assumption:

**Assumption A4 (signal exhaustion).** $\xi$ is the only school-side draw on $\theta^u$. Equivalently, conditional on $(\theta^k, G, Z)$, no other school-side variable (informal parent–teacher communication, sibling outcomes. . . ) is correlated with $\theta^u$. This is the substantive content of treating $Z$ as the school's information set; without it, the recommendation's informational content would be biased.[9]

[9] This is the main limitation of the model so far. Recommendations partly emerge from informal meetings with parents that I do not observe directly, although $\theta_{\text{fam}}^k$ absorbs parental pressure as a baseline type. The implication is that

## 2.5 Equilibrium recommendation rule

In this first version, I do not model the school's strategy structurally[10]. The recommendation $R_{it}$ is the equilibrium outcome of school behavior, conditional on the school's information set and capacity constraints. I treat the equilibrium rule $\rho_\tau$ as a primitive of the institutional environment.

Define $R_{it} \in \mathcal{R} = \mathcal{J} \cup \{\emptyset\}$, $|\mathcal{R}| = 6$. Schools commit to a recommendation rule:

$$\rho_\tau \ : \ (\theta^k, G_t, Z_t, X_t, K_t) \ \longmapsto \ \Delta(\mathcal{R}), \tag{10}$$

indexed by school cycle-architecture type $\tau$. Capacity $K_{s,j,t}$ enters $W_{it}$ as an observable covariate that shifts recommendations under tighter constraints. We can parametrise this as multinomial logit:

$$\Pr\big(R_{it} = r \,\big|\, W_{it};\ \tau, t\big) \ = \ \frac{\exp(W_{it}^\top \beta_{r,\tau,t})}{\sum_{r' \in \mathcal{R}} \exp(W_{it}^\top \beta_{r',\tau,t})}, \tag{11}$$

with $W_{it} = (\theta_i^k, G_{it}, Z_{it}, X_{it}, K_{s,j,t})$.

I impose these two assumptions on the school side:

**Assumption A5 (rule exclusion).**

$$R_{it} \ \perp \ \theta_i^u \ \big|\ (\theta_i^k, G_{it}, Z_{it}, K_{s,j,t}, \tau, t). \tag{12}$$

$\theta^u$ is unobserved by everyone, so the rule cannot condition on it directly. The rule conditions on what the school observes, and the recommendation's informativeness about $\theta^u$ flows entirely through $p(Z \mid \theta^u, \cdot)$. Together with A4, this pins down $R$ as a deterministic function of the school's information set up to the logit shock.

**Assumption A6 (cycle-architecture).** The equilibrium rule depends on schools only through $\tau(s)$. Testable by comparing within-$\tau$ to between-$\tau$ variation in $\hat{\rho}_s$. If within-$\tau$ heterogeneity is large, the CF in §2.8 is reinterpreted as replacing C123 rules with the C1-average rule.

## 2.6 Student's beliefs and Bayesian updating

Let $\mathcal{H}_{it}$ be the student's information at end of period $t$. Within each $t \in \{1, \dots, 4\}$, updating happens in two stages. Here I use the fact that students first observe grades of the standardized exams and that recommendations only come at the end of the year:

**Stage 1: posterior after grades.** Given $\mathcal{H}_{i,t-1} \sim \mathcal{N}(m^H_{i,t-1}, \Sigma^H_{i,t-1})$ and observed grades $G_{it}$, conjugacy gives

$$\theta_i^u \,\big|\, \mathcal{H}_{i,t-1}, G_{it} \ \sim \ \mathcal{N}(m^G_{it}, \Sigma^G_{it}), \tag{13}$$

counterfactuals that change the *information structure* are vulnerable to Lucas critique: if in reality schools and parents keep communicating, a CF that changes these primitives says little about reality. The CFs in §2.8 are chosen to change incentives or what students see, not the school–family communication channel.

[10]This part is still work in progress, it requires another layer of work

with

$$\left(\Sigma_{it}^{G}\right)^{-1} \;=\; \left(\Sigma_{i,t-1}^{H}\right)^{-1} + \sum_{b} \frac{\Lambda_b \Lambda_b^{\top}}{\sigma_b^2}, \tag{14}$$

$$m_{it}^{G} \;=\; \Sigma_{it}^{G}\left[\left(\Sigma_{i,t-1}^{H}\right)^{-1} m_{i,t-1}^{H} + \sum_{b} \frac{\Lambda_b\big(G_{ibt} - \mu_b(\theta_i^k, X_{it})\big)}{\sigma_b^2}\right]. \tag{15}$$

This is a standard result in the literature Arcidiacono et al. (2025); Bunting et al. (2024). Now, the second stage is slightly more tricky —given the nonGaussianity—, I propose the following steps, although this is still preliminary:

**Stage 2: posterior after recommendation.** Given the post-grade posterior and observed $R_{it}$, Bayes gives

$$f(\theta^u \mid \mathcal{H}_{i,t-1}, G_{it}, R_{it}, \tau) \propto \phi(\theta^u; m_{it}^G, \Sigma_{it}^G) \cdot L_i(R_{it} \mid \theta^u; \tau), \tag{16}$$

where the perceived recommendation likelihood, derived under A5 by integrating out $Z$, is

$$L_i(R \mid \theta^u; \tau) = \sum_{m=1}^{M} \rho_\tau(R \mid W_{-Z}, Z = m) \cdot \big[\Phi(\kappa_m - \mu - \lambda^\top \theta^u) - \Phi(\kappa_{m-1} - \mu - \lambda^\top \theta^u)\big]. \tag{17}$$

The integral over the latent $Z^*$ has closed form via $\Phi$; the sum is finite over the ordered-probit cells.

**Assumption A8 (Gaussian moment-matching).** The post-recommendation posterior is approximated as Gaussian by matching the first two moments of the exact mixture-of-truncated-Gaussians posterior. This is the assumed-density-filtering step of Minka (2001); Särkkä (2013) gives the textbook treatment in the filtering literature:

$$\theta_i^u \mid \mathcal{H}_{i,t-1}, G_{it}, R_{it} \overset{\text{approx}}{\sim} \mathcal{N}(m_{it}^R, \Sigma_{it}^R), \tag{18}$$

with closed-form expressions

$$m_{it}^R = \sum_m \pi_m^R \cdot \mathbb{E}_m[\theta^u], \tag{19}$$

$$\Sigma_{it}^R = \sum_m \pi_m^R \big[V_m + (\mathbb{E}_m[\theta^u] - m_{it}^R)(\mathbb{E}_m[\theta^u] - m_{it}^R)^\top\big], \tag{20}$$

where $\pi_m^R \propto \rho_\tau(R \mid Z = m) \cdot \Pr_m$ are the recommendation- weighted cell probabilities, $\mathbb{E}_m[\theta^u]$ and $V_m$ are the conditional mean and variance of $\theta^u$ given $\{Z = m\}$ (closed form via inverse-Mills ratios for doubly-truncated Gaussians), and $\Pr_m$ are the marginal cell probabilities under the post-grade prior. Full derivations in Appendix A.

**Intuition behind this result:** The student doesn't see $Z$, but they see $R$, the school's discrete summary of it. When a student receives ASO-STEM, they reason backward: the school must have seen $Z$ values consistent with that recommendation, given the student's grades. That backward inference is what shifts beliefs.

How much beliefs move depends on the school's rule. If a school recommends ASO-STEM to nearly everyone with similar grades, the recommendation adds little beyond grades and beliefs barely move. If the school reserves ASO-STEM for specific $Z$ patterns (high effort rank, top of class), receiving ASO-STEM is a strong signal and beliefs move sharply. Within tracks, beliefs shift in the direction the recommendation suggests: a high recommendation pulls the posterior mean of $\theta^u$ upward for the recommended track[11].

This is the channel through which institutional incentives translate into student outcomes. A school facing tight capacity may issue ASO-STEM less frequently than a school with the same students but slack capacity. "ASO-STEM" then carries different information at the two schools — not because students learn differently, but because the schools *say* different things in equivalent situations. Students Bayesianly absorb whatever the rule produces; the counterfactuals in §2.8 vary the rule.

## 2.7 State, action, and dynamic choice

The state of student $i$ at the start of $t$ is

$$s_{it} \;=\; \big(\theta_i^k,\; m_{i,t-1}^H,\; \Sigma_{i,t-1}^H,\; d_{i,t-1},\; X_{it}\big). \tag{21}$$

$\Sigma_{i,t-1}^H$ is deterministic given the past sequence of actions (Kalman variance updates do not depend on realized signals), so the effective state is $(\theta_i^k, m_{i,t-1}^H, d_{i,t-1}, X_{it})$. $d_{i,t-1}$ captures the previous track and switching costs.

The action set at $t$ is $\mathcal{A}_t$. At $t=3$, $\mathcal{A}_3 = \mathcal{J}_i \subseteq \mathcal{J}$ restricted by the certificate (A/B/C). For $t \geq 4$, $\mathcal{A}_t$ is the set of feasible downgrades from $d_{i,t-1}$ (BSO absorbing).

Flow utility for $a \in \mathcal{A}_t$:

$$u(s_{it}, a; \theta) \;=\; \alpha_a + \zeta_a^\top \theta_i^k + \delta_{\mathrm{SES}(i)} \cdot e_a^\top m_{i,t-1}^H + \gamma^\top X_{it} + c(d_{i,t-1}, a) + \eta_{ita}, \tag{22}$$

where $\zeta_a$ are track-specific loadings on the discrete type, $e_a$ is the $a$-th unit vector (so $e_a^\top m_{i,t-1}^H$ is the posterior mean of $\theta_{i,a}^u$), $c(\cdot,\cdot)$ captures switching costs, $\eta_{ita} \sim$ T1EV i.i.d., and $\delta_{\mathrm{SES}(i)}$ is an SES-specific weight on belief means[12]. Integration over $\theta^k$ uses EM weights $q_{i,k}$.[13]

The Bellman:

$$V_t(s_{it}) \;=\; \mathbb{E}_\eta \max_{a \in \mathcal{A}_t} \Big\{ u(s_{it}, a; \theta) + \beta\, \mathbb{E}\big[V_{t+1}(s_{i,t+1}) \,\big|\, s_{it}, a\big] \Big\}, \tag{23}$$

with T1EV log-sum:

$$\bar{V}_t(s_{it}) \;=\; \log \sum_{a \in \mathcal{A}_t} \exp(v_t(s_{it}, a)), \qquad v_t(s_{it}, a) \;=\; u(s_{it}, a; \theta) + \beta\, \mathbb{E}\big[\bar{V}_{t+1}(s_{i,t+1}) \,\big|\, s_{it}, a\big]. \tag{24}$$

Choice probabilities:

$$\Pr(d_{it} = a \mid s_{it}) \;=\; \frac{\exp v_t(s_{it}, a)}{\sum_{a' \in \mathcal{A}_t} \exp v_t(s_{it}, a')}. \tag{25}$$

**Assumption A11 (finite dependence).** The downgrade-only dynamics with absorbing BSO yield finite dependence: from any current track $a$, a path of length $K = T - t$ ending in BSO has identical conditional state distribution as the reference path "drop to BSO immediately" after $K$ periods, Arcidiacono and Miller (2011); Arcidiacono et al. (2025) inversion then expresses differenced FV terms in terms of CCPs on the post-recommendation belief state, computed flexibly in Stage 3 of estimation. This avoids backward recursion and full solution methods.

[11] The data supports that students reach on recommendations even conditional on standardized grades results. This supports this approach.

[12] Key heterogeneity in our mdoel would be parental SES and priors. Here baseline beliefs can be made dependent of family priors + how students update can also potentially be a function of SES. Idea is that what matters in all these cases of informational asymmetries is the background: high-SES students likely have more information at home or stronger prior; while low-SES students may be more impacted by the policy.

[13] Note that in this first proposal I have only modeled the effect of recommendation on beliefs. However, there are other channels through which recommendations can affect actions. One is authority; it does not change beliefs but students just follow what schools decide. Including an effect of advice directly on the flow utility —together with the effect on belief— would isolate this. I leave this also for future work.

## 2.8 Counterfactuals

The model has two economic forces that pull in opposite directions. On the one hand, the school knows more than the student: $Z$ contains residual information about $\theta^u$ that students cannot access on their own, so the recommendation is informative. On the other hand, the rule that maps $Z$ into $R$ is shaped by capacity and cycle architecture, so it is also *distorted*. The recommendation is at once a useful signal and a sorting device. Each counterfactual below isolates one side of this tension. I keep the institutional architecture — who talks to whom, how recommendations are produced — fixed, in line with A4 and the partial-equilibrium scope. Some ideas for CF in this setting are:

**CF1: information disclosure.** Show students $Z$ directly while leaving $\rho_\tau$ unchanged. Students update on the school's underlying signal as if it were just another grade; the recommendation $R$ becomes redundant. This isolates the *value of the recommendation as a signal* from its role as a sorting device. The non-obvious result is that disclosure need not help: if the school's rule is well-calibrated, $R$ is already a sufficient summary of $Z$ for tracking purposes, and disclosure mostly adds noise to high-SES students who already have good priors. The students who benefit are those whose families update most aggressively on new information — typically lower-SES, by the $\delta_{\text{SES}}$ channel.

**CF2: capacity-only.** Hold $\hat{\rho}_\tau$ and the rule coefficients $\beta$ fixed; raise C123 capacity to C1 levels —or make schools C123 recommend like C1, this would be cleaner—. The CF measures how much of the welfare gap between school types is attributable to the binding capacity constraint, separately from any difference in how the two types of schools *would* advise under slack capacity. The non-obvious result is that capacity may matter most for the *marginal* student — those whom C123 schools nudge into TSO/BSO to manage cohort sizes — and the welfare effect runs through dynamic skill accumulation rather than through the static track choice.

Ideally, the data also supports modelling the school side structurally. The most policy-relevant counterfactuals — changing the school's objective directly, i.e aligning incentives across cycle architectures or imposing truthful disclosure — require identifying the school's objective $\omega_\tau$, likely from variation in capacity or institutional rules. Fu and Mehta (2018) provides a natural template, but I leave this for future iterations.

# Appendix

## 2.9 Measurements avaialble at baseline

In the following table I map the measures at baseline that I have and how these can be used to recover latent traits:

Table 1: Mapping of observed measures to latent factors

| Latent factor | Observed measures | Source |
|---|---|---|
| $\theta^k_{\text{num}}$ Numeric ability | GETLOV numeric reasoning subscore | Student |
| | Results Math exam School | Student |
| | Spatial IQ test | Student |
| $\theta^k_{\text{verb}}$ Verbal ability | GETLOV verbal reasoning subscore | Student |
| | Results Dutch exam School | Student |
| | Spatial IQ test | Student |
| $\theta^k_{\text{noncog}}$ Non-cognitive skills | PMTK achievement motivation | Student |
| | PMTK task orientation | Student |
| | PMTK positive test anxiety | Student |
| | Big Five subscales (Eindvragenlijst, year 1) | Student |
| $\theta^k_{\text{fam}}$ Family pressure | Mother's education | Parent |
| | Father's education | Parent |
| | Parental homework help | Student |
| | Parental income | Parent |
| | Parental influence in tracking decisions | Student |

*Notes:* Each latent factor is identified from at least three measurements following **?** and Humphries et al. (2025); Arcidiacono et al. (2025). Baseline measurements ($t = 0$) are taken before secondary-school enrollment.

### 2.10 Estimation

Here I sketch a bit how the estimation would go. I think this is just adapting Arcidiacono et al. (2025) with some simplifications and with the added complication of the nonGaussian signal.

**Stage 1 — measurement system + auxiliary regressions → $q_{i,k}$.** Flexible parametric system: (i) baseline measurements $M_i$ for $\theta^k$ identification (Hu–Schennach); (ii) panel auxiliary regressions on grades, recommendations, choices, and the terminal outcome with type dummies. Output: $q_{i,k}$ and $\pi_k$.

**Stage 2 — structural learning parameters.** Weighted EM on the grade equation only, as above. Output: $(\hat{\Lambda}_b, \hat{\sigma}^2_b, \hat{\mu}_b, \hat{\Sigma}_0)$ and $(m^G_{it}, \Sigma^G_{it})$.

**Stage 2.5 — DGP of $Z$ and recommendation rule.** (a) Estimate ordered-probit parameters $(\lambda_b, \kappa_b)$ for $Z$ given Stage-2 posterior moments. (b) Estimate $\beta_{r,\tau,t}$ in (11) by weighted MLE.

**Stage 3 — flexible CCPs and FV terms.** Flexible MNL CCPs on the post-recommendation belief state. Differenced FV terms via finite-dependence path to BSO.

**Stage 4 — structural flow utility.** Weighted MNL on (22) with CCP-derived offset terms. Recovers $(\zeta_a, \delta_{\text{SES}}, \gamma, c, \alpha_a)$.

**Identification of CCPs conditional on type.** Given consistent $q_{i,k}$ from Stage 1, type-conditional CCPs are weighted averages (Arcidiacono and Miller, 2011):

$$\widehat{\Pr}\big(d_{it} = a \mid s_{it} \approx s, \theta^k = k\big) \;=\; \frac{\sum_i q_{i,k}\,\mathbb{1}[d_{it} = a, s_{it} \approx s]}{\sum_i q_{i,k}\,\mathbb{1}[s_{it} \approx s]}.$$

## 2.11 Moment-matched post-recommendation posterior

This appendix states the closed-form expressions for $(m^R_{it}, \Sigma^R_{it})$ in equation (16). The construction follows the assumed-density-filtering approach of Minka (2001) adapted to the ordered-probit signal structure. I drop $i, t$ subscripts.

**Setup.** The post-grade posterior is $\theta^u \sim \mathcal{N}(m^G, \Sigma^G)$. The latent school signal $Z^* = \mu + \lambda^\top \theta^u + \xi$ with $\xi \sim \mathcal{N}(0, 1)$ is observed in $M$ ordered cells via thresholds $\kappa_0 < \kappa_1 < \cdots < \kappa_M$. The exact post-recommendation posterior is a finite mixture of truncated Gaussians, one per $Z$-cell:

$$f(\theta^u \mid R) \;\propto\; \phi(\theta^u; m^G, \Sigma^G) \cdot \sum_{m=1}^{M} \rho_\tau(R \mid Z = m) \cdot \big[\Phi(\kappa_m - \mu - \lambda^\top \theta^u) - \Phi(\kappa_{m-1} - \mu - \lambda^\top \theta^u)\big]. \quad (26)$$

**Matched moments.** Let $\mathbb{E}_m[\theta^u]$ and $V_m$ denote the conditional mean and variance of $\theta^u$ within cell $m$ (closed-form in $\Phi$, $\phi$ via standard truncated-Gaussian formulas). Let $\Pr_m$ be the marginal cell probability under the prior, and define the recommendation-reweighted weights

$$\pi^R_m \;=\; \frac{\rho_\tau(R \mid Z = m) \cdot \Pr_m}{\sum_{m'} \rho_\tau(R \mid Z = m') \cdot \Pr_{m'}}. \quad (27)$$

The first two moments of (26) are

$$m^R = \sum_{m=1}^{M} \pi^R_m \cdot \mathbb{E}_m[\theta^u], \quad (28)$$

$$\Sigma^R = \sum_m \pi^R_m \cdot V_m + \sum_m \pi^R_m \cdot (\mathbb{E}_m[\theta^u] - m^R)(\mathbb{E}_m[\theta^u] - m^R)^\top. \quad (29)$$

The Gaussian moment-matching step (Minka, 2001; Särkkä, 2013) approximates (26) by $\mathcal{N}(m^R, \Sigma^R)$, exact in the first two moments. Validation against numerical quadrature can also be done.

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